

Machine Learning-Based Credit Risk Assessment with Explainable AI

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1 Problem Definition

Credit risk assessment is an important application of machine learning in financial services. The ability to identify customers who may be at higher risk of credit default can support risk assessment and informed decision-making. However, complex machine-learning models may produce accurate predictions without clearly explaining the factors contributing to those predictions.

This project aims to develop a machine learning-based credit risk assessment system with Explainable AI. The system will use historical customer financial behaviour to estimate the probability of future credit default and provide interpretable explanations for the predictions.

The project will use the American Express Default Prediction dataset available through Kaggle. The dataset contains monthly customer-level observations covering areas such as delinquency, spending, payments, balances, and risk-related variables.

The main objective is to develop a complete machine-learning pipeline covering data preprocessing, exploratory analysis, feature engineering, model development, evaluation, and explainability, resulting in a practical credit risk assessment workflow.

2 Data Source

The primary data source will be the American Express Default Prediction dataset available through Kaggle.

The dataset contains longitudinal customer records and a binary default target. Multiple monthly observations are available for customers, allowing historical behavioural information to be incorporated into the analysis.

The dataset contains anonymised variables broadly related to:

- Delinquency
- Spend
- Payment
- Balance
- Risk

Due to the large size of the dataset and applicable data-use conditions, the raw dataset will not be uploaded to the public GitHub repository. Instead, the repository will contain documentation describing the data source and instructions for obtaining the dataset. The raw data will be accessed and processed in the appropriate working environment, while project code, notebooks and results will be maintained in GitHub.

3 Intended Methods

The project will follow an end-to-end machine-learning workflow.

1. **Data preprocessing:** Missing-value analysis, data-type handling, data cleaning, and memory-efficient processing will be performed.
2. **Exploratory Data Analysis:** Customer behaviour and its relationship with default will be investigated using statistical analysis and visualisations.
3. **Feature engineering:** Historical customer observations will be transformed into customer-level features such as recent values, averages, maximum/minimum values, variability, and changes in behaviour over time.
4. **Machine Learning:** Logistic Regression will be used as a baseline model. A tree-based model such as XGBoost or LightGBM will then be investigated as an improved model.
5. **Evaluation:** Models will be evaluated using appropriate metrics such as ROC-AUC, PR-AUC, Precision, Recall, F1-score, and Brier Score. Confusion matrices and other visualisations will also be used.
6. **Explainable AI:** SHAP will be investigated to explain global model behaviour and individual customer predictions.
7. **End-to-end pipeline:** The developed system will demonstrate the complete workflow from customer data preprocessing and feature engineering to credit-risk prediction and explainable model outputs.

The project will apply knowledge from programme modules including Data Science, Advanced Machine Learning, Advanced Data Mining, Advanced AI, Advanced Databases, and Information Visualization.

4 Expected Outputs

The expected outputs are:

1. Cleaned and processed dataset
2. Exploratory data analysis and visualisations
3. Engineered customer behavioural features
4. Logistic Regression baseline model
5. Improved tree-based machine-learning model
6. Comparative model evaluation
7. Default-risk probability predictions
8. SHAP-based model explanations
9. Simple risk-prediction prototype
10. Reproducible Python notebooks and source code
11. Documented GitHub repository.

The final system will demonstrate:

**Customer Data → Preprocessing → Feature Engineering → ML
Prediction → Default Probability → Explanation**

5 Key Risks and Assumptions

- **Large dataset:** The AMEX dataset is very large and may create computational limitations. Memory-efficient and chunk-based processing will therefore be considered.
- **Data leakage:** As customers have multiple monthly observations, customer-level splitting will be used to avoid information from the same customer appearing in both training and validation data.
- **Class imbalance:** Appropriate evaluation metrics will be used rather than relying only on accuracy.
- **Anonymised variables:** Individual feature meanings may be limited. Interpretations will therefore be based on available documentation and observed relationships.
- **Data-use conditions:** The applicable Kaggle competition/data-use conditions will be followed, and raw data will not be redistributed through the public GitHub repository.
- **Scope:** The project will priorities a complete and well-evaluated ML and Explainable AI pipeline rather than attempting to reproduce complex Kaggle leader board solutions.